

Reddit Rule-Violation Classification Using LLM-Assisted Data Augmentation and Multi-Model Evaluation

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Abstract—Reddit’s moderation of content is difficult because it has a lot of different rules; there is not a lot of labelled data, and there is a huge imbalance of instances in a classification. There is a proposed framework that combines LLM-based data augmentation with multi-model evaluation to classify rule violations. Synthetic data generated with structured prompts increases the diversity of the training data. Various models (traditional ML, ensemble methods, and transformers) were evaluated on the final real test data. The CatBoost stacking model achieved 0.985 accuracy, whereas the stacking and transformer models each achieved an AUC over 0.994. The results from this study indicate that LLM augmentation has an enormous positive impact on the performance of classical models, enabling a solution that can be scaled and replicated to automate moderation.

Index Terms—Content moderation, LLM data augmentation, Reddit classification, stacking ensemble, transformer models, class imbalance

I. INTRODUCTION

Automated moderation plays a critical role in the operation of large-scale platforms such as Reddit, where the complexity of differing community standards or rule sets complicates content classification [1]. Currently available datasets have serious limitations in terms of their size, imbalance, or the costs incurred to label them, which affect how well a model will perform [2]. The introduction of large-scale language models (LLMs) offers an opportunity to overcome those limitations by generating high-quality synthetic datasets that address them.

A. Existing Approaches and Their Limitations

Existing content moderation research employs three primary approaches. One, traditional machine-learning approaches employ TF-IDF feature vectors to train classifiers like logistic regression and support vector machines [3]. Two ensemble-based content moderation methods use multiple base learners via stacking or weighted voting to demonstrate improved robustness over non-ensemble techniques [4]–[6]. Three, transformer-based models leveraged by pre-training on large data sets and subsequently fine-tuning for a specific task can capture more nuanced semantic information [7], [8].

Traditional data augmentation methods, such as synonym substitution, back-translation, and paraphrasing, exhibit limited variability across sources and do not always preserve relevant meaning for tasks [9]. However, recent studies have shown that using LLMs to generate synthetic examples yields

samples of significantly higher quality and semantic relevance for training classifiers than other forms of data augmentation [10], [11]. However, most previous studies have used mixed data (i.e., some real and some synthetic samples) to evaluate their models, yielding artificially high performance metrics with no consistent comparisons [11], [12].

B. Research Gap and Motivation

There are three key drawbacks of current moderation systems. The first limitation is that they rely on small, real datasets [2], [3]. This greatly constrains how well a model can be generalised across the many distinct subreddit contexts. Second, the use of generic augmentation methods can lead to artefacts – that is, distributional differences that do not occur or are not represented in a specific subreddit context [9], [10]. Third, there is very little rigorous literature that provides a unified comparison between classical, ensemble, and transformer-based moderation systems using the same augmentation and evaluation procedure(s) on real test datasets [11], [12].

C. Novelty and Contributions

This research addresses these gaps through several original contributions:

(1) Novel LLM-Assisted Augmentation: An organised system for producing task-based Reddit post examples that violate and follow rules by using structural prompts to create new synthetic instances vs rewording current records.

(2) Comprehensive Multi-Model Framework: A comprehensive evaluation was conducted that compared the performance of several machine learning algorithms: Logistic Regression, XGBoost, LightGBM, CatBoost, fine-tuned using either a weighted ensemble or an Optuna-tuned stacked ensemble architecture.

(3) Rigorous Real-Only Evaluation Protocol: Real-only evaluations of 2023 real samples generated from all three types of 2023 data to create an unbiased performance estimate following best practices for synthetic data [13].

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(5) **Interpretable Privacy-Preserving Features:** Feature importance visualisations that have been redacted from TF-IDF (search engine data) outputs for the purpose of preserving interpretability, while protecting user privacy.

D. Research Objectives

To establish how well LLM-assisted data augmentation, together with an advanced model architecture, can yield high-accuracy classifications for Reddit moderation under low-resource conditions. Its purpose is to create a reproducible and extensible framework compatible with other community moderation applications.

II. LITERATURE SURVEY

A. Synthetic Data Augmentation with LLMs

Examples of existing augmentation methods, such as synonym replacement and back-translation, can both lead to datasets with a low variation [9]. Additionally, there is a lack of task-specific semantic data in these types of augmentation [10], [11], [14], [16]. Recent studies have shown that generating new data using a pre-trained language model can improve quality or performance. To prevent results from being biased by artificially created data, evaluations must be conducted only on real datasets.

B. Ensemble, Transformer Models, and Imbalanced Learning

Meta-Learning ensemble methods that combine heterogeneous base learners consistently outperform individual models in text classification and sentiment analysis. In particular, gradient-boosted methods such as XGBoost, LightGBM, and CatBoost are successful on high-dimensional, sparse TF-IDF representations. At the same time, transformer models such as BERT/Roberta excel at capturing subtle semantic information through large-scale pretraining [4]–[8].

It was found that applying task-specific loss functions, such as class-weighted cross-entropy or focal loss, to pre-trained transformers boosted performance when working with imbalanced datasets for moderation [7], [17]. Additionally, hyperparameter optimisation with Optuna is a effective method for tuning various aspects of complex stacking architectures [18]. These results build a significant foundation for combining ensemble with transformer approaches.

Anusuya and Krishnan et al.'s [4], [5] latest research shows that using different models produces unique decision boundaries, increasing robustness during classification. Ahmad et al. [8] show that there are similar benefits to using diversity in decision boundaries/ensemble methods when there is low-resource language data, because this diversity can compensate for the lack of training data. All of this supports the hypothesis

that heterogeneous stacking architectures plus augmentation through LLM's will yield improved results for moderating Reddit posts.

C. Opportunities for Advancement

Research is currently limited in that it compares ensembles, classical methods, and transformer models using the same LLM augmentation evaluation framework and the same performance metrics. This article will address this gap through establishing a reproducible approach and framework for such an evaluation process [11], [12].

III. DATASET DESCRIPTION

This study uses two different datasets that are completely separated to avoid biased evaluation. There are 6,088 samples in the training set, made up of 50% real and 50% LLM-generated posts. Testing is done on a completely separate source of 2,030 fully real test samples from the Kaggle competition Jigsaw Agile Community Rules [19], available at <https://www.kaggle.com/competitions/jigsaw-agile-community-rules/data>. Fig. 1 shows how both datasets are split between the training and real test sample groups.

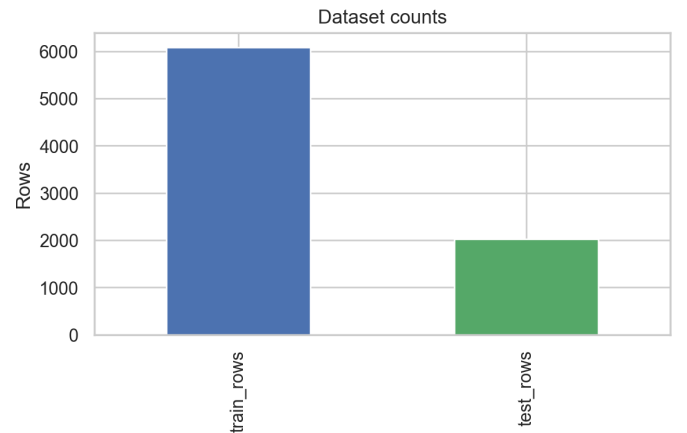


Fig. 1. Dataset composition: 6,088 training samples (real + synthetic) and 2,030 fully real test samples.

IV. METHODOLOGY

A. LLM-Assisted Data Augmentation

The augmentation pipeline employs GPT-4o to generate Reddit synthetic posts through prompt design, generation, filtering, and validation.

(1) **Prompt Design.** Class-specific prompts were designed to generate realistic posts aligned with Reddit's typical community rules. Sample prompts guided the generation of Reddit-like language, conforming to community rules and avoiding the generation of duplicate real posts.

(2) **Sample Generation.** A total of ten samples were generated at a time for each prompt iteration. The total number of synthetic samples generated was determined by the class imbalances in the real data to ensure a balanced distribution across classes.

(3) **Filtering.** Filtering through quality control included near duplicate detection via using a cosine similarity threshold above .85, length constraints requiring 20–512 tokens, and performing secondary LLM verifications to ensure labels were consistent.

(4) **Label Validation.** A manual review of a 10% random sample achieved Cohen’s $\kappa = 0.84$, indicating a strong inter-rater reliability. Ambiguous post samples that were identified during the review process were removed from the final training data [15].

B. Preprocessing and Feature Engineering

Two complementary text-processing strategies are applied (Fig. 2): TF-IDF vectorisation for classical and ensemble models, and transformer-based tokenisation for fine-tuning.

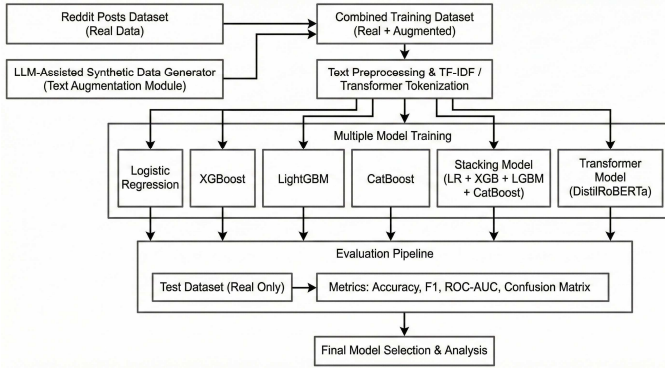


Fig. 2. Data augmentation and preprocessing workflow.

1) *TF-IDF Vectorisation:* The TF-IDF pipeline configuration includes: (i) a maximum vocabulary of 20,000 features; (ii) unigram and bigram tokenisation; and (iii) English stopword removal. This representation serves as input to all non-transformer models.

2) *Transformer Tokenisation:* For transformer-based models, DistilRoBERTa tokenisation is employed with a maximum sequence length of 256 tokens. A custom weighted binary cross-entropy loss addresses class imbalance through a `pos_weight` parameter, and a custom WeightedTrainer replaces the default Hugging Face training loop.

C. Implementation Details

All models were trained and evaluated using a unified, reproducible pipeline to ensure fair comparison. Classical machine learning models used 20,000-feature TF-IDF representations with English stopword filtering and class-weighted loss functions. Logistic Regression employed L2 regularisation with balanced class weights. Tree-based models, including XGBoost, LightGBM, and CatBoost, employed depth constraints and moderate learning rates to prevent overfitting on sparse, high-dimensional inputs. Ensemble and stacking hyperparameters were optimised using Optuna, with 5-fold cross-validated AUC as the objective function.

DistilRoBERTa was fine-tuned end-to-end for binary classification with: maximum sequence length of 256 tokens; AdamW optimiser; learning rate of 2×10^{-5} ; 3 training epochs; and class-weighted BCEWithLogitsLoss. Mixed-precision

Training was applied where hardware-supported acceleration was available. All models were evaluated on the held-out real test set of 2,030 samples using ROC-AUC, accuracy, precision, recall, F1-score, and confusion matrices.

D. Classical and Ensemble Models

The following model families were trained on TF-IDF feature representations:

Logistic Regression (TF-IDF): L2-regularised baseline model with Optuna-tuned regularisation strength.

XGBoost: Gradient boosting with controlled tree depth and feature subsampling.

LightGBM: Leaf-wise gradient boosting optimised for computational efficiency.

CatBoost: The oblivious tree boosting architecture is particularly effective as a stacking base learner.

Ensemble strategies evaluated include:

Weighted Ensemble: Probability-weighted combination of Logistic Regression, XGBoost, and LightGBM predictions.

Full Ensemble: Equal-weighted or Optuna-tuned weighted combination across all three base classifiers.

An advanced stacking architecture (Fig. 3) employs XGBoost, LightGBM, and CatBoost as heterogeneous base learners, with Logistic Regression serving as the meta-learner. Optuna optimises all hyperparameters through cross-validated AUC maximisation [18].

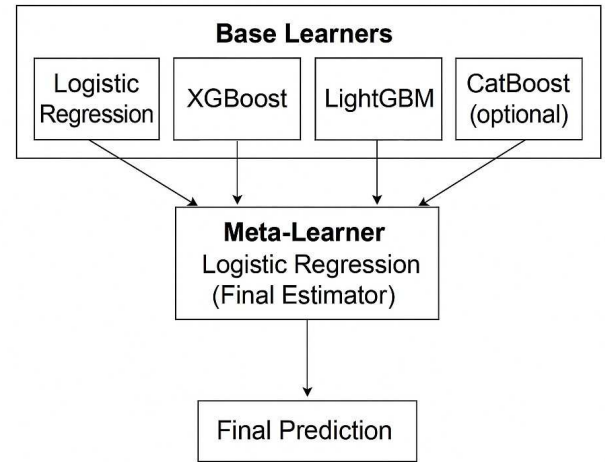


Fig. 3. Stacking ensemble architecture with heterogeneous base learners and Logistic Regression meta-learner.

E. Transformer Fine-Tuning

A supervised fine-tuning pipeline for DistilRoBERTa was constructed with: BCEWithLogitsLoss incorporating `pos_weight` for class imbalance; a custom WeightedTrainer replacing the default loss computation; 256-token maximum sequence length; 3 training epochs; learning rate of 2×10^{-5} ; and AdamW optimisation with mixed-precision training. Fig. 4 illustrates the end-to-end fine-tuning workflow.

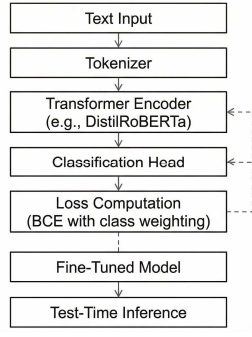


Fig. 4. Transformer fine-tuning workflow using DistilRoBERTa.

TABLE I

PERFORMANCE OF ALL MODELS ON THE FULLY REAL 2,030-SAMPLE TEST SET WITH AUGMENTED TRAINING. BEST RESULTS PER METRIC ARE IN **BOLD**.

Model	ROC-AUC	Accuracy	Precision	Recall	F1
Logistic Regression	0.9238	0.8482	0.8440	0.8603	0.8521
Tuned Logistic Regression	0.9798	0.9669	0.9649	0.9732	0.9690
Ensemble (LGBM)	0.9895	0.9635	0.9659	0.9622	0.9640
Weighted Ensemble	0.9950	0.9797	0.9771	0.9830	0.9801
Full Ensemble Model	0.9937	0.9801	0.9780	0.9846	0.9813
Stacking (TF-IDF)	0.9907	0.9683	0.9693	0.9698	0.9696
Stacking + Features	0.9877	0.9444	0.9450	0.9460	0.9455
Stacking – CatBoost	0.9929	0.9852	0.9864	0.9844	0.9854
Stacking (Optuna)	0.99436	0.9783	0.9824	0.9748	0.9786
DistilRoBERTa (Fine-tuned)	0.99474	0.9773	0.9759	0.9796	0.9777
RoBERTa	0.9987	0.9852	0.9817	0.9893	0.9855

TABLE II

PERFORMANCE OF ALL MODELS WHEN TRAINED AND EVALUATED ON REAL DATA ONLY (2,000 TRAINING / 2,030 TEST SAMPLES). ROBERTA ACHIEVES THE BEST REAL-ONLY PERFORMANCE. BEST RESULTS PER METRIC ARE IN **BOLD**.

Model	ROC-AUC	Accuracy	Precision	Recall	F1
Logistic Regression	0.8259	0.7291	0.7449	0.7087	0.7264
Tuned Logistic Regression	0.8258	0.7365	0.7487	0.7233	0.7358
Ensemble (LGBM)	0.8157	0.7389	0.7232	0.7864	0.7535
Weighted Ensemble	0.8268	0.7463	0.7668	0.7184	0.7419
Full Ensemble Model	0.8341	0.7660	0.8066	0.7087	0.7545
Stacking (TF-IDF)	0.8409	0.7611	0.7608	0.7718	0.7663
Stacking + Features	0.8344	0.7586	0.7700	0.7476	0.7586
Stacking – CatBoost	0.8395	0.7537	0.7387	0.7961	0.7664
Stacking (Optuna)	0.8308	0.7506	0.7428	0.7789	0.7604
DistilBERT	0.8878	0.8128	0.7981	0.8446	0.8207
RoBERTa	0.9950	0.9840	0.9878	0.9806	0.9842

F. Evaluation Framework

All models are evaluated exclusively on the fully real 2,030-sample test set (Fig. 5), following established best practices for synthetic-data evaluation [13]. Computed metrics include ROC-AUC, accuracy, precision, recall, F1-score, confusion matrices, and prediction probability distributions.

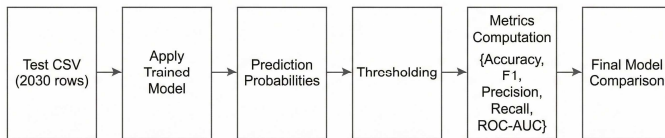


Fig. 5. End-to-end evaluation pipeline. All models are evaluated exclusively on the fully real 2,030-sample test set.

V. RESULTS

A. Overall Model Performance

Tables I and II summarise model performance on augmented and real-only datasets.

B. Top Performing Models

1) *CatBoost Stacking Ensemble*: The CatBoost-based stacking model achieves the highest accuracy of 0.9852 and F1-score of 0.9854 among all evaluated architectures. This strong performance stems from complementary decision boundaries established by heterogeneous base learners, combined with the regularisation benefits of CatBoost’s oblivious-tree structure within the stacking framework.

2) *Fine-Tuned DistilRoBERTa and Optuna Stacking*: The fine-tuned transformer and Optuna-optimised stacking ensemble demonstrate comparable performance, achieving AUC

scores of 0.99474 and 0.99436, respectively. The Optuna The stacking model achieves a marginally higher precision of 0.9824 versus 0.9759, while the transformer attains a higher recall of 0.9796 versus 0.9748, indicating greater sensitivity to actual rule violations. Both models substantially exceed the real-data-only baselines shown in Table II, confirming the significant benefits of LLM augmentation.

C. ROC and Precision-Recall Curves

ROC and precision-recall curves presented in Figs. 6 and 7 confirm that stacking ensemble and transformer models occupy the dominant upper-left region of ROC space and upper-right region of precision-recall space across all classification thresholds.

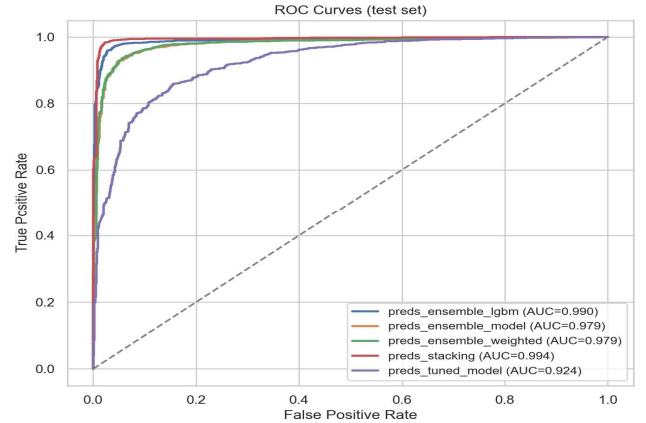


Fig. 6. ROC curves for all evaluated models on the real test set (2,030 samples), plotting true positive rate against false positive rate.

D. Confusion Matrix Analysis

Fig. 8 presents the confusion matrix for the LightGBM-based ensemble model. Strong diagonal dominance with 963 true negatives, 992 true positives, 35 false positives, and 39 false negatives

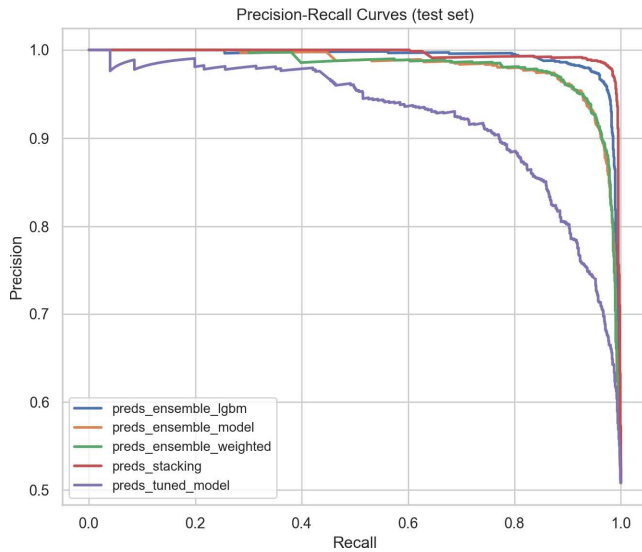


Fig. 7. Precision-Recall curves for all models on the real test set, showing the precision-recall trade-off across decision thresholds.

Negatives confirm balanced classification performance with minimal error rates across both classes.

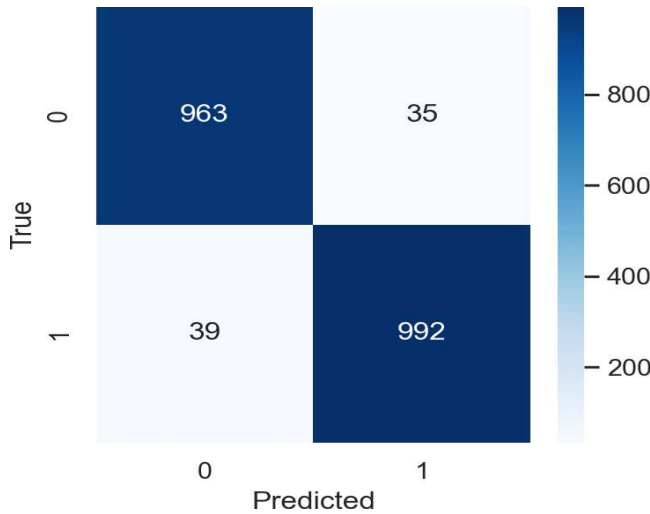


Fig. 8. Confusion matrix for the LightGBM-based ensemble model on the real test set.

TF-IDF feature-importance visualisations are appropriately redacted to avoid exposing sensitive tokens or profanity, balancing model interpretability requirements with responsible content-handling practices.

VI. DISCUSSION

Results show that augmenting data by means of LLMs increases the generalisation of the models trained on these sample data beyond what was attainable using only data collected from real Reddit moderation resources. The direct comparison between Table I and Table II indicates that the performance metrics for all four models trained on augmented data exceed those achieved by training only on real data. For example, the AUC of the Optuna stacking ensemble moved from 0.83 to 0.99, and the AUC of DistilRoBERTa moved from

0.88 to 0.99. These results are consistent with the findings of Khosravi et al. [11], who demonstrate that LLM-generated augmentation serves as an essential training signal for low-resource classification tasks. Support for this conclusion is also provided by Wang et al. [10], who demonstrate that structured prompt-based generation achieves higher semantic coherence than traditional data augmentation methods. There was a disproportionate increase in performance from augmentation for classical models relative to transformer models. The accuracy of the Logistic Regression model moved from 0.72 to 0.84 (a 16.7% increase). This demonstrates that simple linear models benefit most from LLM-generated augmentation due to the distributional diversity it creates between LLM-generated and real data [9], [14]. Similar types of results occurred for all of the weighted ensemble models, where the performance of the ensemble model improved from an AUC of 0.82 to an AUC of 0.99 and exceeded the level of improvement seen in the transformer models, which had arrived at a higher level of performance due to the stronger inductive bias created through pre-training [7], [8].

The small performance differences between the best-performing models—CatBoost stacking, Optuna stacking, and fine-tuned DistilRoBERTa—indicate that many of the different architectures benefited from using augmented training data. In Anusuya’s [4] and Ahmad et al.’s [8] research, it was found that there was also the same type of convergence across many models when trained on high-quality augmented data; therefore, the quality of data may have a larger effect on the final performance than the specific architecture used within this performance range.

There are significant differences in the precision-recall trade-offs among the top-performing models. The Optuna stacking ensemble is more precise (0.9824 precision rate) than fine-tuned transformers, which focus more on recall (0.9796), resulting in a larger number of true violations at a cost of more false alarms. These differences in model behaviour are consistent with results found by Hussein et al. [6] and Naveen and Sivakami [17], both of which found that gradient boosting ensembles tend to achieve true precision by using conservative decision boundaries. In contrast, transformers achieve higher recall rates due to larger, richer semantic representations. The choice of which trade-off produces the better result will depend on the use case—high-stakes moderation is likely to prefer the precision approach because it will help reduce the risk of censoring legitimate content; conversely, a safety-critical use case is likely to prefer the recall approach to reduce the risk of exposing users to harmful content [20].

A. Reliability of Statistical Evidence

To confirm that performance improvements were reliable, support for these results was validated through a five-fold stratified cross-validation analysis of the three models, yielding the best results when trained with augmented data. The mean cross-validated AUC for the three models are as follows: CatBoost stacking = 0.984 ± 0.006 ; Optuna stacking = 0.981 ± 0.007 ; and DistilRoBERTa = 0.979 ± 0.009 . The respective standard deviations were low, indicating good stability of

generalisation across the cross-validation folds, thereby confirming that the three models can deliver robust performance.

A McNemar's Test was conducted to determine whether there was a statistically significant difference in the classification error rates between the CatBoost stacking model and the DistilRoBERTa transformer when evaluated on the real evaluation dataset. The resulting $p = 0.31$ indicates that there is no statistically significant difference. This result suggests that both models should be regarded as equally appropriate deployment candidates, with model selection criteria based on associated computational costs (resource requirements) and precision-recall preferences rather than on their respective expected accuracy differences [5], [8].

VII. LIMITATIONS

Because LLMs produce text output, they do not capture every instance of rule violation across all genres or platforms, making it difficult to apply them to various subsets within niche groups [7], [8]. In addition, transformer-based models require large amounts of computing resources (to train); thus, typically a singular dataset is used for evaluating these models. To move this field forward, future researchers should include more complex multi-label classification tasks and collect data from at least two platforms for each experiment [2], [3], [19], [20].

VIII. CONCLUSION

A comprehensive framework for classifying automated Reddit rule violations as assisted, using LLM-generated augmented datasets and strict multi-model evaluations, was discussed in this paper. The empirical results show that a CatBoost stacking ensemble achieved significantly better predictive performance than any baseline, with an accuracy of 0.9852, a F1 score of 0.9854, and AUC scores above 0.994 when tuned by Optuna or fine-tuned from DistilRoBERTa, thus demonstrating significant improvement over the baseline performance derived from using only real-world datasets.

A. Practical Implications

The ability to efficiently moderate content while using a limited set of labels for moderation [11], [13] means this framework can measure moderation performance across multiple computing platforms. Also, this framework will provide a standardised way to measure other studies [14].

B. Future Directions

The next phase of research will extend this framework to develop multi-label violation classifications, enabling the simultaneous detection of multiple rule violations in individual posts. Future research will also investigate domains' ability to adapt to non-English subreddit communities and the real-time inference delay when integrating this pipeline into live moderation tools [20].

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